pyAmpliCol at one loop

Benchmarks of the recursive one-loop engine against MadLoop, Recola and OpenLoops

report 2026-11-11 · newest engine measured `524e6cfa` · finite part, binary64, one pinned thread, $B = 128$
Table 1: engine `524e6cfa`, G1 2026-11-11, AMD EPYC 9754, shared host · Table 2: engine `524e6cfa`

×1/16 faster — ×1/4 — ×1 — ×4 — ×16 slower
run time ÷ fastest provider

Gate pass ×0.25 · fail ×1.4 · open ×1.0 · <5 ms: not gated ×1.4 — time ÷ limit
Stages inputs · numerator · scatter · NinjaRS · masters · glue
missing (why) · no colour: generation, LC · † within estimate · ¶ reference's own (DD) · ‡ user decision
ML MadLoop · Rc Recola · OL OpenLoops

Table 1: **Born-interfered one-loop processes** (the one-loop amplitude interfered with the Born). How to read it: at the end.

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
	pyAmpliCol: the most recent measured configuration — `loop` · $2 \to 2$ to $2 \to 5$: engine `524e6cfaa78cc150fc40e2e5f7381626f199390a`, native `loop-native-v2-slice-678c4749 (9de26a2e…)`, G1 2026-11-11 · AMD EPYC 9754, shared host			
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at $B = 128$, the gate batch; provider times; gate G1 (time ÷ limit); stage shares	batched, same configuration; LC/FC; no provider ratio
$2 \to 2$	$gg \to gg$ 4 legs · QCD · 24 families (LC 12) Rc·OL ≤ 3.0e−13 · ML 1.3e−12 16 pts · median 2.4e−14 · †	**9.32 s** ×36 0.57 GiB peak · plan 0.0240 MB **Rc 258 ms** · ML 9.70 s	**1.44 ms** ×0.68 **ML 2.13 ms** · OL 7.61 ms · Rc 18.9 ms $B = 128$ no gate (2→2) numerator 43 %	**0.909 ms** LC/FC 0.63 $B = 128$ · FC 1.44 ms
$2 \to 3$	$u\bar{u} \to d\bar{d}g$ 5 legs · QCD · 42 families (LC 27) ML·OL·Rc ≤ 3.0e−12 16 pts · median 6.9e−14	**10.4 s** ×13 0.58 GiB peak · plan 0.0300 MB **Rc 783 ms** · ML 9.87 s	**3.02 ms** ×0.84 **OL 3.60 ms** · Rc 5.86 ms · ML 5.93 ms $B = 128$ <5 ms: not gated ×0.64 numerator 34 %	**2.03 ms** LC/FC 0.67 $B = 128$ · FC 3.02 ms
	$u\bar{u} \to u\bar{u}g$ 5 legs · QCD · 68 families (LC 42) OL·ML·Rc ≤ 6.0e−12 16 pts · median 1.7e−13	**13.7 s** ×27 0.58 GiB peak · plan 0.0360 MB **Rc 503 ms** · ML 10.9 s	**2.43 ms** ×0.64 **OL 3.82 ms** · Rc 5.41 ms · ML 9.03 ms $B = 128$ <5 ms: not gated ×0.49 numerator 38 %	**1.50 ms** LC/FC 0.62 $B = 128$ · FC 2.43 ms
	$u\bar{u} \to e^-e^+g$ 5 legs · QCD · 11 families (LC 7) ML·Rc ≤ 7.5e−14 16 pts · median 1.4e−14	**17.4 s** ×21 0.61 GiB peak · plan 0.0530 MB **Rc 815 ms** · ML 14.0 s	**0.672 ms** ×1.1 **Rc 0.593 ms** · ML 0.945 ms · OL no library $B = 128$ <5 ms: not gated ×0.87 numerator 34 %	**0.499 ms** LC/FC 0.74 $B = 128$ · FC 0.672 ms
	$u\bar{u} \to ggg$ 5 legs · QCD · 90 families (LC 62) Rc·ML ≤ 9.2e−13 · OL 8.4e−11 16 pts · median 5.2e−14 · ¶	**11.8 s** ×11 0.58 GiB peak · plan 0.0490 MB **Rc 1.06 s** · ML 17.6 s	**6.73 ms** ×0.30 **OL 22.4 ms** · Rc 46.0 ms · ML 52.2 ms $B = 128$ pass ×0.23 numerator 43 %	**3.18 ms** LC/FC 0.47 $B = 128$ · FC 6.73 ms
	$gg \to ggg$ 5 legs · QCD · 154 families (LC 77) Rc 1.4e−12 · OL·ML ≤ 3.0e−6 16 pts · median 5.8e−13 · †	**12.3 s** ×1.1 0.60 GiB peak · plan 0.112 MB **Rc 11.7 s** · ML 47.0 s	**24.6 ms** ×0.14 **OL 171 ms** · ML 287 ms · Rc 784 ms $B = 128$ pass ×0.11 numerator 40 %	**11.3 ms** LC/FC 0.46 $B = 128$ · FC 24.6 ms
$2 \to 4$	$u\bar{u} \to e^-e^+gg$ 6 legs · QCD · 69 families (LC 33) Rc 3.7e−13 · ML 1.9e−10 16 pts · median 7.2e−14 · †	**19.6 s** ×13 0.62 GiB peak · plan 0.0680 MB **Rc 1.48 s** · ML 24.8 s	**6.14 ms** ×0.33 **Rc 18.9 ms** · ML 61.3 ms · OL no library $B = 128$ pass ×0.25 numerator 48 %	**2.72 ms** LC/FC 0.44 $B = 128$ · FC 6.14 ms
	$gg \to t\bar{t}gg$ 6 legs · QCD · 1 086 families (LC 375) Rc 1.3e−11 · OL·ML ≤ 6.6e−9 16 pts · median 1.2e−11 · †	**19.3 s** ×0.30 0.64 GiB peak · plan 0.320 MB **Rc 63.9 s** · ML 218 s	**76.1 ms** ×0.15 **OL 501 ms** · Rc 2.19 s · ML 5.65 s $B = 128$ pass ×0.12 numerator 37 %	**28.1 ms** LC/FC 0.37 $B = 128$ · FC 76.1 ms
	$u\bar{u} \to gggg$ 6 legs · QCD · 628 families (LC 375) OL·Rc ≤ 2.8e−10 16 pts · median 1.7e−12 · †	**23.4 s** ×0.65 0.66 GiB peak · plan 0.382 MB **Rc 35.8 s**	**58.7 ms** ×0.15 **OL 398 ms** · Rc 1.23 s · ML no production timing $B = 128$ pass ×0.11 numerator 48 %	**26.7 ms** LC/FC 0.46 $B = 128$ · FC 58.7 ms
	$gg \to gggg$ 6 legs · QCD · 1 100 families (LC 550) Rc·OL·ML ≤ 1.7e−7 8 pts · median 2.3e−11 · †	**20.2 s** ×0.0095 0.90 GiB peak · plan 1.23 MB **Rc 35.4 min** · ML 144.6 min	**463 ms** ×0.097 **OL 4.80 s** · Rc 17.6 s^{p} · ML 56.9 s $B = 128$ pass ×0.074 inputs 42 %	**175 ms** LC/FC 0.38 $B = 128$ · FC 463 ms

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
	pyAmpliCol: the most recent measured configuration — loop · $2 \to 2$ to $2 \to 5$: engine 524e6cfaa78cc150fc40e2e5f7381626f199390a, native loop-native-v2-slice-678c4749 (9de26a2e…), G1 2026-11-11 · AMD EPYC 9754, shared host			
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at $B = 128$, the gate batch; provider times; gate G1 (time ÷ limit); stage shares	batched, same configuration; LC/FC; no provider ratio
$2 \to 5$	$u\bar{u} \to e^-e^+d\bar{d}g$ 7 legs · QCD · 242 families (LC 114) OL·ML·Rc ≤ 4.5e−11 16 pts · median 8.9e−13	**14.8 s** ×2.1 0.63 GiB peak · plan 0.0960 MB **Rc 6.95 s** · ML 58.6 s	**14.3 ms** ×0.37 $B = 128$ **Rc 38.2 ms** · OL 76.4 ms · ML 167 ms pass ×0.29 · numerator 43 %	**6.51 ms** LC/FC **0.46** $B = 128$ · FC 14.3 ms
	$u\bar{u} \to e^-e^+ggg$ 7 legs · QCD · 463 families (LC 164) Rc·OL·ML ≤ 6.1e−11 16 pts · median 2.3e−12 · †	**14.8 s** ×1.1 0.64 GiB peak · plan 0.179 MB **Rc 13.9 s** · ML 146 s	**43.2 ms** ×0.15 $B = 128$ **Rc 286 ms** · OL 363 ms · ML 1.48 s pass ×0.12 · numerator 58 %	**9.38 ms** LC/FC **0.22** $B = 128$ · FC 43.2 ms
	$gg \to t\bar{t}e^-e^+g$ 7 legs · QCD · 579 families (LC 164) Rc 2.6e−10 · OL·ML ≤ 1.7e−8 16 pts · median 1.7e−11 · †	**20.7 s** ×0.64 0.65 GiB peak · plan 0.176 MB **Rc 32.3 s** · ML 165 s	**75.4 ms** ×0.10 $B = 128$ **OL 726 ms** · Rc 854 ms · ML 4.02 s pass ×0.080 numerator 48 %	**15.5 ms** LC/FC **0.21** $B = 128$ · FC 75.4 ms
	$gg \to t\bar{t}e^-e^+g$ 7 legs · EW · 1 599 families OL 2.7e−10 · ML 2.8e−9 16 pts · median 7.6e−10	**453 s** ×13 9.40 GiB peak · plan 0.445 MB **Rc 34.2 s** · ML 11.9 min	**181 ms** ×0.12 $B = 128$ **OL 1.52 s** · Rc 2.09 s · ML 11.3 s pass ×0.091 numerator 39 % model: the workspace's py3 G_mu copy	***not supported*** refused by design: EW LC/NLC is never supported (user 2026-10-30, Q4; typed colour-declaration-coverage)
	$u\bar{u} \to ggggg$ 7 legs · QCD · 5 258 families (LC 2 712) ML 2.1e−12 · Rc 2.9e−10 3 pts · median 7.0e−11 · ¶	**136 s** ×0.013 2.76 GiB peak · plan 14.7 MB **Rc 170.3 min** · ML 12.2 h	**1.81 s** ≥×0.0099 $B = 128$ **Rc ≤182 s** · ML 569 s[p] · OL no library pass ×0.0032 numerator 52 %	**596 ms** LC/FC **0.33** $B = 128$ · FC 1.81 s
	$gg \to ggggg$ 7 legs · QCD · 4 707 families (LC 4 707) NJet DD 1.5e−7 10 pts · median 3.4e−9 · † · top-free (particle_veto = [t]), Nf = 5 ‡ pts 4, 8 above 2.0e−8: accepted (user 2026-10-25)	**173 s** 4.39 GiB peak · plan 20.2 MB ML diagrams 17.4 h, stopped in export · Rc unfinished	**5.11 s** $B = 128$ no gate (no matched provider) inputs 46 %	**4.01 s** LC/FC **0.78** $B = 128$ · FC 5.11 s
	$2 \to 6$: pyAmpliCol only, no provider exists at $2 \to 6$, so no ratio, no gate and no colour · engine 2f3f76d9 (work/v2-mf-8g-3), native loop-native-mf-2to6-2f3f76d9-lto (2bae17f9…), an ancestor of the engine above, not re-run on it · shared host, indicative · internal checks only			
$2 \to 6$	$gg \to gggggg$ 8 legs · QCD · 47 348 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 1.0e−11 · E ≤ 1.4e−3 (2 pts above 10^{-5}) · binary64 = DD to 7.4e−5 · rescue 11/16 · NGluon DD to 6.3e−13 · **FC** ε^{-2} ≤ 8.4e−11 · E ≤ 5.1e−3 (2 pts above 10^{-5}) · Catani ε^{-1} ≤ 7.1e−5 · binary64 = DD to 1.2e−4 · rescue 11/16 Ward FC not run, Ward LC not asserted	**55.3 min** FC 74.0 GiB peak · plan 486 MB · store 25.8 min artifact 0.53 GB · load 4.21 s (1.06 GiB) **LC** 46.1 min · 63.9 GiB · artifact 0.364 GB	**201 s** 1 block of 16 $B = 16$ no provider: no gate inputs 55 % peak 40.1 GiB	**98.0 s** 1 block of 128 $B = 128$ numerator 54 % peak 27.2 GiB
	$u\bar{u} \to gggggg$ 8 legs · QCD · 47 348 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 4.6e−12 · E ≤ 6.2e−7 · binary64 = DD to 8.4e−11 · rescue 1/16 · Ward ≤ 6.5e−12 · **FC** ε^{-2} ≤ 1.1e−11 · E ≤ 5.9e−6 · Catani ε^{-1} ≤ 3.3e−8 · binary64 = DD to 1.7e−10 · rescue 2/16 · Ward ≤ 2.1e−11	**73.6 min** FC 92.9 GiB peak · plan 814 MB · store 449 s artifact 0.143 GB · load 3.85 s (1.66 GiB) **LC** 20.5 min · 47.0 GiB · artifact 0.047 GB	**62.1 s** 1 block of 128 $B = 128$ no provider: no gate numerator 77 % peak 31.7 GiB	**10.9 s** 1 block of 128 $B = 128$ numerator 60 % peak 12.6 GiB
	$u\bar{u} \to u\bar{u}gggg$ 8 legs · QCD · 40 454 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 6.7e−14 · E ≤ 7.5e−7 · binary64 = DD to 1.3e−11 · rescue 2/16 · Ward ≤ 4.3e−12 · **FC** ε^{-2} ≤ 2.0e−13 · E ≤ 6.5e−6 · Catani ε^{-1} ≤ 6.6e−11 · binary64 = DD to 1.3e−11 · rescue 4/16 · Ward ≤ 8.5e−12 (DD 1.1e−28)	**29.8 min** FC 42.5 GiB peak · plan 315 MB · store 46.4 s artifact 0.038 GB · load 1.51 s (0.68 GiB) **LC** 506 s · 18.2 GiB · artifact 0.009 GB	**19.2 s** 1 block of 128 $B = 128$ no provider: no gate numerator 75 % peak 36.0 GiB	**2.67 s** 1 block of 128 $B = 128$ numerator 55 % peak 7.38 GiB

Table 2: **Loop-induced processes** ($|A_{\text{loop}}|^2$; the equivalent of Table 1), timed with the unchecked reduction setting[1]. How to read it: at the end.

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked
pyAmpliCol: engine 524e6cfa, native loop-native-v2-slice-678c4749 (9de26a2e…) on itphlies (packages landed: lit-amp, lit-r2, lit-ew, lit-eft) · cells: TIRust with the unchecked setting (= auto with it: shown paired on gg_hgg), NinjaRS unchecked beside gate on TIRust · measured 2026-11-11 · host state: up to 41 rows timed in parallel units, 1-min load 203–630, MemAvailable ≥ 247.9 GiB, CPU share of the timed calls ≥ 0.838 (median per row), 17 rows re-timed once on another core (a disturbed first sample, or a stream regenerated; named in their records) · nothing estimated **Protocol:** reduction_unchecked = true (default false) (Runtime.load / evaluate keyword; config evaluation.reduction_unchecked), explicit tirust and ninjars; poles="none", no stability check, no rescue; $B = 128$ per call (median over rounds); one thread pinned to one core (taskset, RAYON_NUM_THREADS=1); the 128 distinct stream points; 3 fresh runtimes per arm in rotated order, the median of their medians (spread ×: largest/smallest); providers F4 most optimistic, timed in the same unit on the same core, alternating with pyAmpliCol; engine 524e6cfa replication: docs/performance_reports/loop/README.md (unchecked Table 2)		
legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + filter + plan export; optional probes apart (carrier); peak memory; provider generation	median per point at $B = 128$ (fresh runtimes, spread); providers paired (most optimistic); gate or rule on TIRust; stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue (instrumented run: stage shares, not the timed call)
$2 \to 1 \cdot 1 \to 2$		
$gg \to h$ 3 legs · QCD · LI-A1 AN 2.6e−13 · OL 6.2e−13 · ML 9.2e−13 · Rc 9.3e−13 · pass · 6 pts	**12.4 s** cold 17.1 s · 0.63 GiB peak warm 0.61 GiB peak · plan 0.0100 MB	**0.345 ms** ×4.9 **Rc 0.0705 ms** · ML 0.0726 ms · OL 0.177 ms $B = 128$ · 3 runtimes ×1.0 no gate (2→1) **TIRust** (tirust) glue 85 % vs 256-bit truth (128 pts): max 1.2e−13 **NinjaRS per amplitude** (ninjars) 0.363 ms ×5.2 glue 80 % vs 256-bit truth (128 pts): max 1.2e−12
$h \to gg$ 3 legs · QCD · LI-A2 AN 6.2e−14 · Rc 6.3e−13 · ML 6.3e−13 · pass · 3 pts	**9.62 s** cold 16.4 s · 0.63 GiB peak warm 0.62 GiB peak · plan 0.0100 MB	**0.368 ms** ×5.0 **Rc 0.0731 ms** · ML 0.0805 ms · OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries $B = 128$ · 3 runtimes ×1.1 no gate (1→2) **TIRust** (tirust) glue 84 % vs 256-bit truth (128 pts): max 1.0e−13 **NinjaRS per amplitude** (ninjars) 0.386 ms ×5.3 glue 80 % vs 256-bit truth (128 pts): max 1.1e−12
$h \to aa$ 3 legs · QED · LI-A3, LI-E1 AN 2.9e−14 · ML 1.9e−13 · Rc 2.0e−13 · pass · 3 pts	**114 s** cold 190 s · 3.32 GiB peak warm 2.71 GiB peak · plan 0.00400 MB	**0.668 ms** ×6.8 **Rc 0.0975 ms** · ML 0.157 ms · OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries $B = 128$ · 3 runtimes ×1.0 no gate (1→2) **TIRust** (tirust) inputs 70 % vs 256-bit truth (128 pts): max 2.4e−14 **NinjaRS per amplitude** (ninjars) 0.717 ms ×7.4 inputs 74 % vs 256-bit truth (128 pts): max 4.2e−13
$h \to za$ 3 legs · QED · LI-A4, LI-E1 ML 1.6e−14 · AN 1.7e−14 · Rc 7.9e−14 · pass · 3 pts	**154 s** cold 231 s · 3.50 GiB peak warm 2.98 GiB peak · plan 0.00400 MB	**1.000 ms** ×4.5 **Rc 0.220 ms** · ML 0.274 ms · OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries $B = 128$ · 3 runtimes ×1.0 no gate (1→2) **TIRust** (tirust) inputs 68 % vs 256-bit truth (128 pts): max 1.2e−14 **NinjaRS per amplitude** (ninjars) 1.06 ms ×4.8 inputs 72 % vs 256-bit truth (128 pts): max 1.6e−12
$aa \to h$ 3 legs · QED · LI-E2 AN 5.2e−14 · ML 2.2e−13 · Rc 2.2e−13 · pass · 3 pts	**159 s** cold 232 s · 3.32 GiB peak warm 2.73 GiB peak · plan 0.00400 MB	**0.708 ms** ×6.2 **Rc 0.114 ms** · ML 0.157 ms · OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) $B = 128$ · 3 runtimes ×1.0 no gate (2→1) **TIRust** (tirust) inputs 73 % vs 256-bit truth (128 pts): max 1.4e−14 **NinjaRS per amplitude** (ninjars) 0.742 ms ×6.5 inputs 74 % vs 256-bit truth (128 pts): max 1.6e−13
$2 \to 2$		
$gg \to aa$ 4 legs · QCD · LI-A5 ML 3.4e−14 · Rc 4.5e−14 · AN 4.6e−14 · OL 4.7e−14 · pass · 6 pts	**19.2 s** cold 35.8 s · 0.70 GiB peak warm 0.69 GiB peak · plan 0.0210 MB	**1.13 ms** ×0.59 **ML 1.90 ms** · OL 2.65 ms · Rc 3.99 ms $B = 128$ · 3 runtimes ×1.4 no gate (2→2) **TIRust** (tirust) numerator 47 % vs 256-bit truth (128 pts): max 6.6e−13 **NinjaRS per amplitude** (ninjars) 3.54 ms ×1.9 NinjaRS 54 % vs 256-bit truth (128 pts): max 2.4e−13
$gg \to zz$ 4 legs · QCD · LI-21 Rc 4.6e−12 · OL 5.9e−12 · ML 6.6e−12 · pass · 3 pts	**19.1 s** cold 29.6 s · 0.72 GiB peak warm 0.70 GiB peak · plan 0.0290 MB	**2.36 ms** ×0.73 **ML 3.25 ms** · OL 4.12 ms · Rc 5.46 ms $B = 128$ · 3 runtimes ×1.4 no gate (2→2) **TIRust** (tirust) numerator 52 % vs 256-bit truth (128 pts): max 5.5e−12 **NinjaRS per amplitude** (ninjars) 6.10 ms ×1.9 NinjaRS 61 % vs 256-bit truth (128 pts): max 8.8e−11

[1]The setting is off by default. A default call (reduction="auto", checks of the backends on) runs TIRust at the target accuracy (10^{-5}) with the per-point NinjaRS fallback; it does not get these times. Its served fractions and checked times are lit-perf's records in data/li-timing.json.

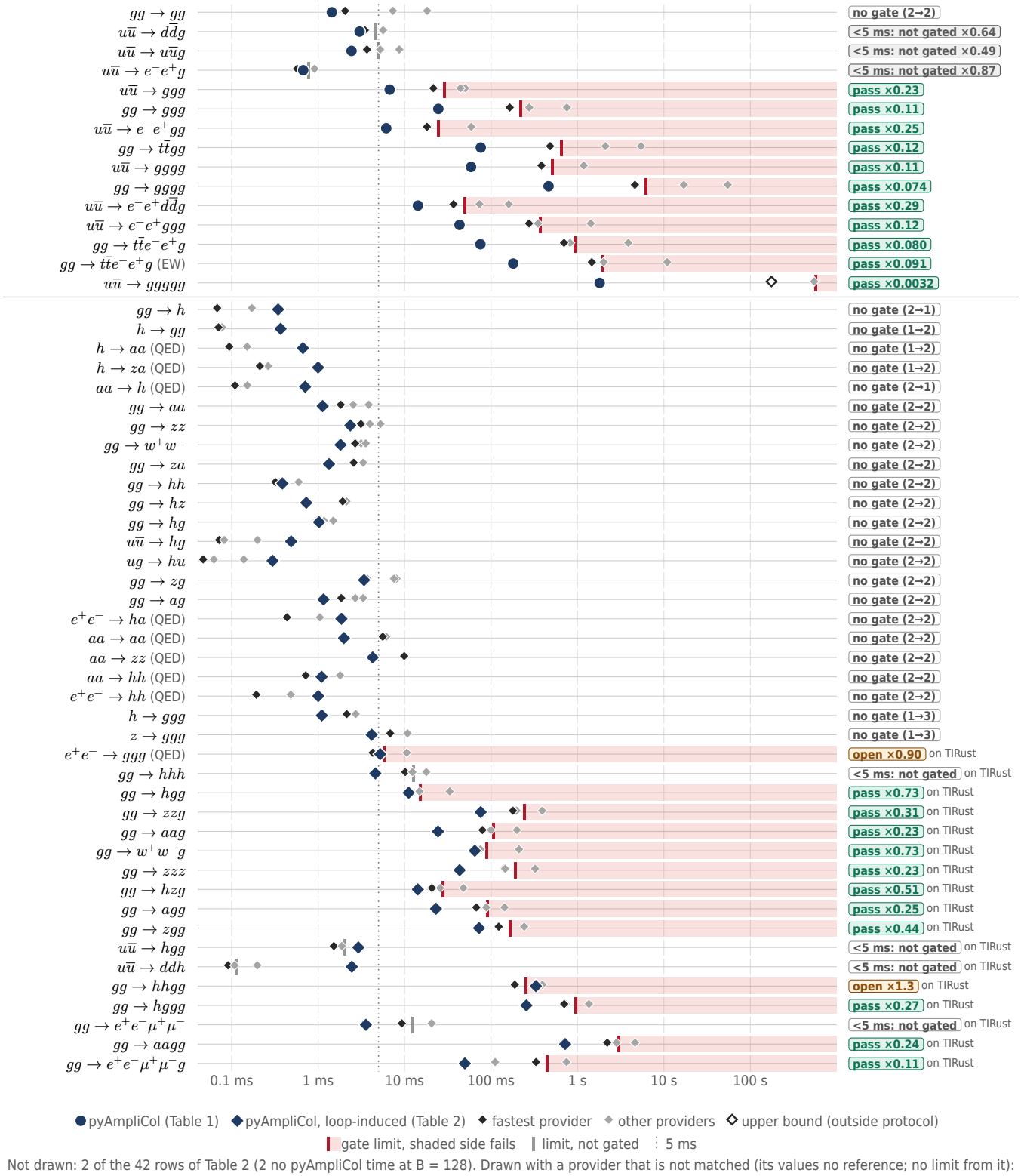
Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked
pyAmpliCol: engine 524e6cfa, native loop-native-v2-slice-678c4749 (9de26a2e…) on itphlies (packages landed: lit-amp, lit-r2, lit-ew, lit-eft) · cells: TIRust with the unchecked setting (= auto with it: shown paired on gg_hgg), NinjaRS unchecked beside gate on TIRust · measured 2026-11-11 · host state: up to 41 rows timed in parallel units, 1-min load 203–630, MemAvailable ≥ 247.9 GiB, CPU share of the timed calls ≥ 0.838 (median per row), 17 rows re-timed once on another core (a disturbed first sample, or a stream regenerated; named in their records) · nothing estimated **Protocol:** reduction_unchecked = true (default false) (Runtime.load / evaluate keyword; config evaluation.reduction_unchecked), explicit tirust and ninjars; poles="none", no stability check, no rescue; $B = 128$ per call (median over rounds); one thread pinned to one core (taskset, RAYON_NUM_THREADS=1); the 128 distinct stream points; 3 fresh runtimes per arm in rotated order, the median of their medians (spread ×: largest/smallest); providers F4 most optimistic, timed in the same unit on the same core, alternating with pyAmpliCol; engine 524e6cfa replication: docs/performance_reports/loop/README.md (unchecked Table 2)		
legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + filter + plan export; optional probes apart (carrier); peak memory; provider generation	median per point at $B = 128$ (fresh runtimes, spread); providers paired (most optimistic); gate or rule on TIRust; stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue (instrumented run: stage shares, not the timed call)
$2 \to 2$		
$gg \to w^+w^-$ 4 legs · QCD · LI-22 OL 1.7e−13 · Rc 5.7e−13 · ML 7.8e−13 · pass · 3 pts	**15.3 s** cold 19.5 s · 0.79 GiB peak warm 0.79 GiB peak · plan 0.0420 MB	**1.82 ms** ×0.65 **OL 2.79 ms** · ML 3.27 ms · Rc 3.70 ms $B = 128$ · 3 runtimes ×1.8 no gate (2→2) **TIRust** (tirust) numerator 46 % vs 256-bit truth (128 pts): max 1.2e−12 **NinjaRS per amplitude** (ninjars) 8.06 ms ×2.9 NinjaRS 66 % vs 256-bit truth (128 pts): max 5.2e−12
$gg \to za$ 4 legs · QCD · LI-23 Rc 1.7e−14 · OL 1.9e−14 · ML 4.2e−14 · pass · 3 pts	**10.9 s** cold 16.9 s · 0.71 GiB peak warm 0.70 GiB peak · plan 0.0360 MB	**1.34 ms** ×0.50 **OL 2.68 ms** · ML 3.38 ms · Rc 3.45 ms $B = 128$ · 3 runtimes ×1.1 no gate (2→2) **TIRust** (tirust) numerator 49 % vs 256-bit truth (128 pts): max 6.8e−13 **NinjaRS per amplitude** (ninjars) 3.82 ms ×1.4 NinjaRS 60 % vs 256-bit truth (128 pts): max 6.4e−12
$gg \to hh$ 4 legs · QCD · LI-24, LI-A7 OL 3.2e−14 · Rc 9.3e−14 · ML 1.6e−13 · pass · 3 pts	**11.7 s** ×119 cold 27.0 s · 0.64 GiB peak warm 0.63 GiB peak · plan 0.0100 MB Rc 98.4 ms	**0.387 ms** ×1.2 **ML 0.332 ms** · Rc 0.373 ms · OL 0.619 ms $B = 128$ · 3 runtimes ×1.7 no gate (2→2) **TIRust** (tirust) glue 48 % vs 256-bit truth (128 pts): max 3.4e−9 **NinjaRS per amplitude** (ninjars) 0.564 ms ×1.7 NinjaRS 41 % vs 256-bit truth (128 pts): max 8.8e−11
$gg \to hz$ 4 legs · QCD · LI-25 ML 5.5e−14 · Rc 6.0e−14 · OL 2.0e−13 · pass · 3 pts	**30.0 s** cold 36.6 s · 0.71 GiB peak warm 0.71 GiB peak · plan 0.0280 MB	**0.729 ms** ×0.37 **Rc 2.00 ms** · ML 2.19 ms · OL 2.21 ms $B = 128$ · 3 runtimes ×1.4 no gate (2→2) **TIRust** (tirust) glue 37 % vs 256-bit truth (128 pts): max 1.4e−12 **NinjaRS per amplitude** (ninjars) 1.26 ms ×0.63 NinjaRS 35 % vs 256-bit truth (128 pts): max 1.4e−12
$gg \to hg$ 4 legs · QCD · LI-26, LI-A7 OL 5.3e−15 · ML 7.1e−15 · Rc 1.3e−14 · pass · 3 pts	**12.7 s** cold 27.3 s · 0.65 GiB peak warm 0.63 GiB peak · plan 0.0110 MB	**1.02 ms** ×1.0 **OL 1.01 ms** · Rc 1.21 ms · ML 1.55 ms $B = 128$ · 3 runtimes ×3.1 no gate (2→2) **TIRust** (tirust) glue 44 % vs 256-bit truth (128 pts): max 2.4e−14 **NinjaRS per amplitude** (ninjars) 1.66 ms ×1.6 NinjaRS 44 % vs 256-bit truth (128 pts): max 7.3e−13
$u\bar{u} \to hg$ 4 legs · QCD · LI-26, LI-A7 Rc 2.0e−15 · ML 3.9e−15 · OL 5.1e−15 · pass · 3 pts	**12.2 s** cold 28.8 s · 0.64 GiB peak warm 0.63 GiB peak · plan 0.0100 MB	**0.487 ms** ×6.5 **ML 0.0744 ms** · Rc 0.0852 ms · OL 0.206 ms $B = 128$ · 3 runtimes ×1.0 no gate (2→2) **TIRust** (tirust) glue 86 % vs 256-bit truth (128 pts): max 1.1e−14 **NinjaRS per amplitude** (ninjars) 0.532 ms ×7.2 glue 81 % vs 256-bit truth (128 pts): max 1.5e−13
$ug \to hu$ 4 legs · QCD · LI-26 OL 5.0e−15 · ML 5.3e−15 · Rc 9.4e−15 · pass · 3 pts	**11.3 s** cold 17.4 s · 0.64 GiB peak warm 0.63 GiB peak · plan 0.0100 MB	**0.297 ms** ×6.1 **ML 0.0485 ms** · Rc 0.0645 ms · OL 0.144 ms $B = 128$ · 3 runtimes ×1.1 no gate (2→2) **TIRust** (tirust) glue 84 % vs 256-bit truth (128 pts): max 1.2e−14 **NinjaRS per amplitude** (ninjars) 0.315 ms ×6.5 glue 84 % vs 256-bit truth (128 pts): max 1.1e−13
$gg \to zg$ 4 legs · QCD · LI-27 Rc 4.3e−14 · OL 1.1e−13 · ML 1.9e−13 · pass · 3 pts	**16.1 s** cold 34.4 s · 0.71 GiB peak warm 0.71 GiB peak · plan 0.0290 MB	**3.40 ms** ×0.90 **OL 3.79 ms** · Rc 7.83 ms · ML 8.38 ms $B = 128$ · 3 runtimes ×1.3 no gate (2→2) **TIRust** (tirust) glue 42 % vs 256-bit truth (128 pts): max 9.4e−13 **NinjaRS per amplitude** (ninjars) 12.9 ms ×3.4 NinjaRS 66 % vs 256-bit truth (128 pts): max 9.7e−12
$gg \to ag$ 4 legs · QCD · LI-27 OL 1.6e−13 · ML 1.9e−13 · Rc 1.9e−13 · pass · 3 pts	**13.5 s** cold 19.7 s · 0.71 GiB peak warm 0.70 GiB peak · plan 0.0230 MB	**1.15 ms** ×0.60 **ML 1.92 ms** · OL 2.80 ms · Rc 3.45 ms $B = 128$ · 3 runtimes ×2.2 no gate (2→2) **TIRust** (tirust) numerator 47 % vs 256-bit truth (128 pts): max 2.3e−13 **NinjaRS per amplitude** (ninjars) 3.09 ms ×1.6 NinjaRS 65 % vs 256-bit truth (128 pts): max 1.7e−12

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked
pyAmpliCol: engine 524e6cfa, native loop-native-v2-slice-678c4749 (9de26a2e…) on itphlies (packages landed: lit-amp, lit-r2, lit-ew, lit-eft) · cells: TIRust with the unchecked setting (= auto with it: shown paired on gg_hgg), NinjaRS unchecked beside gate on TIRust · measured 2026-11-11 · host state: up to 41 rows timed in parallel units, 1-min load 203–630, MemAvailable ≥ 247.9 GiB, CPU share of the timed calls ≥ 0.838 (median per row), 17 rows re-timed once on another core (a disturbed first sample, or a stream regenerated; named in their records) · nothing estimated **Protocol:** reduction_unchecked = true (default false) (Runtime.load / evaluate keyword; config evaluation.reduction_unchecked), explicit tirust and ninjars; poles="none", no stability check, no rescue; $B = 128$ per call (median over rounds); one thread pinned to one core (taskset, RAYON_NUM_THREADS=1); the 128 distinct stream points; 3 fresh runtimes per arm in rotated order, the median of their medians (spread ×: largest/smallest); providers F4 most optimistic, timed in the same unit on the same core, alternating with pyAmpliCol; engine 524e6cfa replication: docs/performance_reports/loop/README.md (unchecked Table 2)		
legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + filter + plan export; optional probes apart (carrier); peak memory; provider generation	median per point at $B = 128$ (fresh runtimes, spread); providers paired (most optimistic); gate or rule on TIRust; stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue (instrumented run: stage shares, not the timed call)
$2 \to 2$		
$e^+e^- \to ha$ 4 legs · QED · LI-A4, LI-E3 Rc 1.2e−14 · ML 1.3e−14 · pass · 3 pts	**363 s** cold 463 s · 5.01 GiB peak warm 4.40 GiB peak · plan 0.00500 MB	**1.86 ms** **×4.1** **Rc 0.453 ms** · ML 1.09 ms · OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) $B = 128$ · 3 runtimes ×1.0 no gate (2→2) **TIRust** (tirust) inputs 73 % vs 256-bit truth (128 pts): max 3.8e−14 **NinjaRS per amplitude** (ninjars) 1.98 ms ×4.4 inputs 75 % vs 256-bit truth (128 pts): max 1.9e−13
$aa \to aa$ 4 legs · QED · LI-A6, LI-E2 Rc 1.5e−14 · ML 1.5e−14 · AN 2.0e−14 · pass · 3 pts	**255 s** cold 309 s · 4.95 GiB peak warm 4.53 GiB peak · plan 0.00500 MB	**1.99 ms** **×0.34** **Rc 5.81 ms** · ML 6.37 ms · OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) $B = 128$ · 3 runtimes ×1.8 no gate (2→2) **TIRust** (tirust) inputs 63 % vs 256-bit truth (128 pts): max 1.8e−14 **NinjaRS per amplitude** (ninjars) 6.78 ms ×1.2 inputs 62 % vs 256-bit truth (128 pts): max 4.9e−14
$aa \to zz$ 4 legs · QED · LI-E2 ML 2.5e−14 · Rc 2.6e−14 · pass · 3 pts	**210 s** cold 290 s · 5.21 GiB peak warm 4.84 GiB peak · plan 0.00500 MB	**4.28 ms** **×0.42** **Rc 10.3 ms** · OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML not timed: no production timing yet $B = 128$ · 3 runtimes ×1.4 no gate (2→2) **TIRust** (tirust) inputs 58 % vs 256-bit truth (128 pts): max 1.5e−13 **NinjaRS per amplitude** (ninjars) 11.8 ms ×1.1 inputs 57 % vs 256-bit truth (128 pts): max 7.4e−13
$aa \to hh$ 4 legs · QED · LI-E2 Rc 6.1e−13 · ML 6.1e−13 · pass · 3 pts	**132 s** cold 200 s · 3.69 GiB peak warm 3.00 GiB peak · plan 0.00400 MB	**1.10 ms** **×1.5** **Rc 0.746 ms** · ML 1.87 ms · OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) $B = 128$ · 3 runtimes ×1.0 no gate (2→2) **TIRust** (tirust) inputs 74 % vs 256-bit truth (128 pts): max 8.8e−11 **NinjaRS per amplitude** (ninjars) 1.55 ms ×2.1 inputs 74 % vs 256-bit truth (128 pts): max 9.1e−11
$e^+e^- \to hh$ 4 legs · QED · LI-E3 Rc 1.3e−10 · ML 1.3e−10 · pass · 3 pts	**229 s** cold 321 s · 4.73 GiB peak warm 4.32 GiB peak · plan 0.00400 MB	**1.00 ms** **×5.0** **Rc 0.200 ms** · ML 0.501 ms · OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) $B = 128$ · 3 runtimes ×1.0 no gate (2→2) **TIRust** (tirust) inputs 81 % vs 256-bit truth (128 pts): max 1.3e−10 **NinjaRS per amplitude** (ninjars) 1.02 ms ×5.1 inputs 80 % vs 256-bit truth (128 pts): max 1.3e−10
$1 \to 3$		
$h \to ggg$ 4 legs · QCD · LI-37 Rc 7.9e−13 · ML 7.0e−12 · partial · 3 pts	**9.11 s** cold 21.3 s · 0.64 GiB peak warm 0.63 GiB peak · plan 0.0110 MB	**1.10 ms** **×0.50** **Rc 2.22 ms** · ML 2.82 ms · OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries $B = 128$ · 3 runtimes ×4.1 no gate (1→3) **TIRust** (tirust) glue 40 % vs 256-bit truth (128 pts): max 6.7e−13 **NinjaRS per amplitude** (ninjars) 1.33 ms ×0.60 NinjaRS 44 % vs 256-bit truth (128 pts): max 1.5e−12
$z \to ggg$ 4 legs · QCD · LI-37 ML 7.8e−12 · Rc 3.2e−11 · partial · 3 pts	**18.9 s** cold 35.7 s · 0.71 GiB peak warm 0.71 GiB peak · plan 0.0290 MB	**4.17 ms** **×0.59** **Rc 7.09 ms** · ML 11.2 ms · OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries $B = 128$ · 3 runtimes ×1.0 no gate (1→3) **TIRust** (tirust) numerator 47 % vs 256-bit truth (128 pts): max 5.5e−11 **NinjaRS per amplitude** (ninjars) 11.1 ms ×1.6 NinjaRS 64 % vs 256-bit truth (128 pts): max 3.8e−10

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked
pyAmpliCol: engine 524e6cfa, native loop-native-v2-slice-678c4749 (9de26a2e…) on itphlies (packages landed: lit-amp, lit-r2, lit-ew, lit-eft) · cells: TIRust with the unchecked setting (= auto with it: shown paired on gg_hgg), NinjaRS unchecked beside gate on TIRust · measured 2026-11-11 · host state: up to 41 rows timed in parallel units, 1-min load 203–630, MemAvailable ≥ 247.9 GiB, CPU share of the timed calls ≥ 0.838 (median per row), 17 rows re-timed once on another core (a disturbed first sample, or a stream regenerated; named in their records) · nothing estimated **Protocol:** reduction_unchecked = true (default false) (Runtime.load / evaluate keyword; config evaluation.reduction_unchecked), explicit tirust and ninjars; poles="none", no stability check, no rescue; $B = 128$ per call (median over rounds); one thread pinned to one core (taskset, RAYON_NUM_THREADS=1); the 128 distinct stream points; 3 fresh runtimes per arm in rotated order, the median of their medians (spread ×: largest/smallest); providers F4 most optimistic, timed in the same unit on the same core, alternating with pyAmpliCol; engine 524e6cfa replication: docs/performance_reports/loop/README.md (unchecked Table 2)		
legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + filter + plan export; optional probes apart (carrier); peak memory; provider generation	median per point at $B = 128$ (fresh runtimes, spread); providers paired (most optimistic); gate or rule on TIRust; stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue (instrumented run: stage shares, not the timed call)
$2 \to 3$		
$e^+e^- \to ggg$ 5 legs · QED · LI-E4 ML 1.1e−10 · Rc 1.4e−8 · partial · 3 pts	**94.3 s** ×517 cold 125 s · 2.81 GiB peak warm 2.66 GiB peak · plan 0.0380 MB Rc 182 ms	**5.21 ms** ×1.2 **Rc 4.44 ms** · ML 11.0 ms · OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) $B = 128$ · 3 runtimes ×1.4 open ×0.90 on TIRust **TIRust** (tirust) glue 41 % vs 256-bit truth (128 pts): max 3.5e−7 **NinjaRS per amplitude** (ninjars) 9.98 ms ×2.2 NinjaRS 59 % vs 256-bit truth (128 pts): max 3.2e−10
$gg \to hhh$ 5 legs · QCD · LI-31 ML 2.8e−13 · OL 3.2e−13 · Rc 8.1e−12 · pass · 3 pts	**10.5 s** ×92 cold 16.7 s · 0.65 GiB peak warm 0.63 GiB peak · plan 0.0120 MB Rc 114 ms	**4.60 ms** ×0.43 **Rc 10.6 ms** · ML 12.8 ms · OL 18.5 ms $B = 128$ · 3 runtimes ×1.0 <5 ms: not gated on TIRust **TIRust** (tirust) TIR moments 65 % vs 256-bit truth (128 pts): max 7.6e−7 **NinjaRS per amplitude** (ninjars) 5.36 ms ×0.51 NinjaRS 57 % **accuracy: 2/128 pts above 1.0e−5, worst 4.5e64, finite values where the checked mode refuses**
$gg \to hgg$ 5 legs · QCD · LI-32, LI-A7 OL 2.7e−10 · ML 2.7e−10 · Rc 2.8e−10 · partial · 3 pts	**13.2 s** ×71 cold 19.1 s · 0.75 GiB peak warm 0.74 GiB peak · plan 0.0190 MB Rc 187 ms	**11.2 ms** ×0.95 **OL 11.7 ms** · Rc 15.4 ms · ML 34.5 ms $B = 128$ · 3 runtimes ×1.1 pass ×0.73 on TIRust **TIRust** (tirust) numerator 43 % vs 256-bit truth (128 pts): max 9.2e−10 **NinjaRS per amplitude** (ninjars) 22.5 ms ×1.9 NinjaRS 63 % vs 256-bit truth (128 pts): max 8.0e−10 auto with the setting: values bitwise TIRust's, 0 per-point fallbacks, time ×1.00
$gg \to zzg$ 5 legs · QCD · LI-33 ML 2.0e−8 · Rc 2.0e−8 · OL 8.1e−8 · partial · 3 pts	**33.5 s** ×31 cold 51.4 s · 1.52 GiB peak warm 1.52 GiB peak · plan 0.0380 MB Rc 1.07 s	**76.0 ms** ×0.41 **Rc 186 ms** · OL 205 ms · ML 406 ms $B = 128$ · 3 runtimes ×1.1 pass ×0.31 on TIRust **TIRust** (tirust) numerator 45 % **accuracy: 18/128 pts above 1.0e−5, worst 1.2e−2, all served by the gate-0 PV last resort** **NinjaRS per amplitude** (ninjars) 460 ms ×2.5 NinjaRS 82 % vs 256-bit truth (128 pts): max 5.0e−8
$gg \to aag$ 5 legs · QCD · LI-33 ML 3.3e−11 · Rc 3.3e−11 · OL 4.8e−11 · partial · 3 pts	**36.2 s** ×115 cold 42.0 s · 0.96 GiB peak warm 0.95 GiB peak · plan 0.0300 MB Rc 314 ms	**24.4 ms** ×0.29 **OL 82.7 ms** · Rc 104 ms · ML 207 ms $B = 128$ · 3 runtimes ×1.3 pass ×0.23 on TIRust **TIRust** (tirust) numerator 58 % vs 256-bit truth (128 pts): max 7.0e−6 **NinjaRS per amplitude** (ninjars) 170 ms ×2.1 NinjaRS 79 % **accuracy: 1/128 pts above 1.0e−5, worst 1.0e−3, also in the checked mode**
$gg \to w^+w^-g$ 5 legs · QCD · LI-33 ML 8.6e−10 · Rc 3.9e−9 · OL 4.3e−8 · partial · 3 pts	**45.3 s** ×30 cold 52.2 s · 2.21 GiB peak warm 2.21 GiB peak · plan 0.0580 MB Rc 1.51 s	**65.0 ms** ×0.94 **OL 68.8 ms** · Rc 78.2 ms · ML 219 ms $B = 128$ · 3 runtimes ×1.1 pass ×0.73 on TIRust **TIRust** (tirust) TIR moments 43 % **accuracy: 24/128 pts above 1.0e−5, worst 1.4e−2, all served by the gate-0 PV last resort** **NinjaRS per amplitude** (ninjars) 320 ms ×4.7 NinjaRS 81 % vs 256-bit truth (128 pts): max 6.2e−8
$gg \to zzz$ 5 legs · QCD · LI-34 ML 9.4e−12 · OL 2.8e−11 · Rc 2.6e−10 · partial · 3 pts	**23.3 s** ×25 cold 29.8 s · 1.05 GiB peak warm 1.04 GiB peak · plan 0.0330 MB Rc 937 ms	**43.3 ms** ×0.29 **OL 147 ms** · Rc 151 ms · ML 336 ms $B = 128$ · 3 runtimes ×1.1 pass ×0.23 on TIRust **TIRust** (tirust) numerator 68 % vs 256-bit truth (128 pts): max 2.3e−8 **NinjaRS per amplitude** (ninjars) 20/128 failed closed, lane status Unstable NinjaRS 77 % **accuracy: 20 pts fail (Unstable), the call timed whole**
$gg \to hzg$ 5 legs · QCD · LI-35 ML 1.9e−13 · Rc 2.1e−13 · OL 2.5e−13 · pass · 3 pts	**13.1 s** ×53 cold 29.8 s · 0.76 GiB peak warm 0.75 GiB peak · plan 0.0330 MB Rc 246 ms	**14.2 ms** ×0.66 **Rc 21.5 ms** · OL 26.7 ms · ML 49.7 ms $B = 128$ · 3 runtimes ×1.2 pass ×0.51 on TIRust **TIRust** (tirust) numerator 54 % vs 256-bit truth (128 pts): max 8.2e−9 **NinjaRS per amplitude** (ninjars) 2/128 failed closed, lane status Unstable, lane status MasterFailed NinjaRS 69 % **accuracy: 2 pts fail (MasterFailed, Unstable), the call timed whole**

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked
pyAmpliCol: engine 524e6cfa, native loop-native-v2-slice-678c4749 (9de26a2e…) on itphlies (packages landed: lit-amp, lit-r2, lit-ew, lit-eft) · cells: TIRust with the unchecked setting (= auto with it: shown paired on gg_hgg), NinjaRS unchecked beside gate on TIRust · measured 2026-11-11 · host state: up to 41 rows timed in parallel units, 1-min load 203–630, MemAvailable ≥ 247.9 GiB, CPU share of the timed calls ≥ 0.838 (median per row), 17 rows re-timed once on another core (a disturbed first sample, or a stream regenerated; named in their records) · nothing estimated **Protocol:** reduction_unchecked = true (default false) (Runtime.load / evaluate keyword; config evaluation.reduction_unchecked), explicit tirust and ninjars; poles="none", no stability check, no rescue; $B = 128$ per call (median over rounds); one thread pinned to one core (taskset, RAYON_NUM_THREADS=1); the 128 distinct stream points; 3 fresh runtimes per arm in rotated order, the median of their medians (spread ×: largest/smallest); providers F4 most optimistic, timed in the same unit on the same core, alternating with pyAmpliCol; engine 524e6cfa replication: docs/performance_reports/loop/README.md (unchecked Table 2)		
legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + filter + plan export; optional probes apart (carrier); peak memory; provider generation	median per point at $B = 128$ (fresh runtimes, spread); providers paired (most optimistic); gate or rule on TIRust; stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue (instrumented run: stage shares, not the timed call)
$2 \to 3$		
$gg \to agg$ 5 legs · QCD · LI-35 Rc 1.3e−11 · ML 2.1e−11 · OL 3.9e−11 · partial · 3 pts	**32.0 s** ×38 cold 44.4 s · 2.08 GiB peak warm 2.08 GiB peak · plan 0.0460 MB Rc 836 ms	**23.1 ms** ×0.33 **OL 70.2 ms** · Rc 91.5 ms · ML 149 ms — $B = 128$ · 3 runtimes ×1.1 pass ×0.25 on TIRust **TIRust** (tirust) numerator 54 % vs 256-bit truth (128 pts): max 3.3e−6 **NinjaRS per amplitude** (ninjars) 98.3 ms ×1.4 NinjaRS 78 % **accuracy: 1/128 pts above 1.0e−5, worst 2.0e−4, also in the checked mode**
$gg \to zgg$ 5 legs · QCD · LI-35 Rc 2.9e−6 · ML 2.9e−6 · partial · 3 pts	**23.6 s** ×14 cold 30.1 s · 2.18 GiB peak warm 2.17 GiB peak · plan 0.0520 MB Rc 1.72 s	**72.7 ms** ×0.57 **Rc 127 ms** · ML 251 ms · OL unavailable: no OpenLoops ls library serves g g > z g g: the public channel database of OpenLoops 2.1.5 (proclib/channels_public.rinfo, 2026-11-06) has zgg/zggg (ppvj2) but no zgggg channel; row ML/RC only (PLAN §3.2) — $B = 128$ · 3 runtimes ×1.1 pass ×0.44 on TIRust **TIRust** (tirust) numerator 41 % **accuracy: 3/128 pts above 1.0e−5, worst 3.1e−5, all served by the gate-0 PV last resort** **NinjaRS per amplitude** (ninjars) 330 ms ×2.6 NinjaRS 78 % vs 256-bit truth (128 pts): max 6.2e−6
$u\bar{u} \to hgg$ 5 legs · QCD · LI-36 Rc 7.7e−10 · ML 7.8e−10 · OL 7.8e−10 · partial · 3 pts	**12.1 s** ×13 cold 17.3 s · 0.65 GiB peak warm 0.63 GiB peak · plan 0.0110 MB Rc 960 ms	**2.92 ms** ×1.9 **OL 1.58 ms** · Rc 1.95 ms · ML 3.08 ms — $B = 128$ · 3 runtimes ×1.5 <5 ms: not gated on TIRust **TIRust** (tirust) TIR moments 40 % vs 256-bit truth (128 pts): max 3.7e−9 **NinjaRS per amplitude** (ninjars) 2.39 ms ×1.5 NinjaRS 47 % vs 256-bit truth (128 pts): max 3.5e−8
$u\bar{u} \to d\bar{d}h$ 5 legs · QCD · LI-36 OL 8.1e−11 · Rc 1.2e−10 · ML 1.2e−10 · pass · 3 pts	**12.1 s** ×41 cold 17.5 s · 0.65 GiB peak warm 0.63 GiB peak · plan 0.0100 MB Rc 293 ms	**2.46 ms** ×26 **Rc 0.0944 ms** · ML 0.112 ms · OL 0.205 ms — $B = 128$ · 3 runtimes ×3.1 <5 ms: not gated on TIRust **TIRust** (tirust) glue 91 % vs 256-bit truth (128 pts): max 4.1e−10 **NinjaRS per amplitude** (ninjars) 0.991 ms ×11 glue 89 % vs 256-bit truth (128 pts): max 4.3e−9
$2 \to 4$		
$gg \to hhgg$ 6 legs · QCD · LI-41 Rc 2.1e−9 · ML 2.8e−9 · OL 1.2e−7 · partial · 4 pts	**55.0 s** ×140 cold 64.3 s · 2.26 GiB peak warm 2.23 GiB peak · plan 0.0470 MB Rc 393 ms	**331 ms** ×1.7 **Rc 196 ms** · OL 346 ms · ML 408 ms — $B = 128$ · 3 runtimes ×1.1 open ×1.3 on TIRust **TIRust** (tirust) TIR moments 78 % vs 256-bit truth (32 pts): max 7.1e−7 **NinjaRS per amplitude** (ninjars) 600 ms ×3.1 NinjaRS 87 % vs 256-bit truth (32 pts): max 7.0e−7
$gg \to hggg$ 6 legs · QCD · LI-42 Rc 2.2e−11 · ML 3.0e−11 · OL 4.2e−11 · partial · 3 pts	**111 s** ×32 cold 111 s · 2.93 GiB peak warm 2.93 GiB peak · plan 0.100 MB Rc 3.46 s	**257 ms** ×0.35 **Rc 730 ms** · OL 759 ms · ML 1.41 s — $B = 128$ · 3 runtimes ×1.2 pass ×0.27 on TIRust **TIRust** (tirust) numerator 43 % vs 256-bit truth (32 pts): max 5.9e−8 **NinjaRS per amplitude** (ninjars) 950 ms ×1.3 NinjaRS 75 % **accuracy: 2/32 pts above 1.0e−5, worst 4.2e−5, also in the checked mode**
$gg \to e^+e^-\mu^+\mu^-$ 6 legs · QCD · LI-43 OL 2.7e−13 · ML 3.4e−13 · pass · 3 pts	**26.4 s** ×97 cold 40.4 s · 0.78 GiB peak warm 0.78 GiB peak · plan 0.0380 MB Rc 271 ms	**3.57 ms** ×0.37 Rc 3.61 ms · **OL 9.66 ms** · ML 21.3 ms — $B = 128$ · 3 runtimes ×1.0 Rc not matched (values excluded): ×0.99 to it <5 ms: not gated on TIRust **TIRust** (tirust) glue 38 % vs 256-bit truth (32 pts): max 5.4e−12 **NinjaRS per amplitude** (ninjars) 5.58 ms ×0.58 NinjaRS 40 % vs 256-bit truth (32 pts): max 6.5e−10

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked
pyAmpliCol: engine 524e6cfa, native loop-native-v2-slice-678c4749 (9de26a2e…) on itphlies (packages landed: lit-amp, lit-r2, lit-ew, lit-eft) · cells: TIRust with the unchecked setting (= auto with it: shown paired on gg_hgg), NinjaRS unchecked beside gate on TIRust · measured 2026-11-11 · host state: up to 41 rows timed in parallel units, 1-min load 203–630, MemAvailable ≥ 247.9 GiB, CPU share of the timed calls ≥ 0.838 (median per row), 17 rows re-timed once on another core (a disturbed first sample, or a stream regenerated; named in their records) · nothing estimated **Protocol:** reduction_unchecked = true (default false) (Runtime.load / evaluate keyword; config evaluation.reduction_unchecked), explicit tirust and ninjars; poles="none", no stability check, no rescue; $B = 128$ per call (median over rounds); one thread pinned to one core (taskset, RAYON_NUM_THREADS=1); the 128 distinct stream points; 3 fresh runtimes per arm in rotated order, the median of their medians (spread ×: largest/smallest); providers F4 most optimistic, timed in the same unit on the same core, alternating with pyAmpliCol; engine 524e6cfa replication: docs/performance_reports/loop/README.md (unchecked Table 2)		
legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + filter + plan export; optional probes apart (carrier); peak memory; provider generation	median per point at $B = 128$ (fresh runtimes, spread); providers paired (most optimistic); gate or rule on TIRust; stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue (instrumented run: stage shares, not the timed call)
$2 \to 4$		
$gg \to aagg$ 6 legs · QCD · LI-44 Rc 6.8e−6 · OL 6.8e−6 · partial · 3 pts	**233 s** ×26 cold 220 s · 3.81 GiB peak warm 3.80 GiB peak · plan 0.112 MB Rc 8.83 s	**720 ms** ×0.31 **OL 2.30 s** · Rc 2.93 s · ML 4.81 s $B = 128$ · 3 runtimes ×1.2 pass ×0.24 on TIRust **TIRust** (tirust) numerator 45 % **TIRust vs NinjaRS (128 pts; no 256-bit truth within the row budget): max 7.0e−2, 23 pts above 1.0e−5; vs 256-bit truth at its 2 largest-difference pts: TIRust 1.3e−4, NinjaRS 7.6e−2** **NinjaRS per amplitude** (ninjars) 5.50 s ×2.4 no instrumented run **TIRust vs NinjaRS (128 pts; no 256-bit truth within the row budget): max 7.0e−2, 23 pts above 1.0e−5; vs 256-bit truth at its 2 largest-difference pts: TIRust 1.3e−4, NinjaRS 7.6e−2**
$gg \to zzzz$ 6 legs · QCD · LI-44 Rc 5.2e−6 · partial · 3 pts	**236 s** ×8.7 cold 238 s · 3.83 GiB peak warm 3.78 GiB peak · plan 0.0630 MB Rc 27.0 s	***failed closed*** tirust (TIRust): 4/128 failed closed, lane status TIRust Unstable **Rc 6.19 s** · ML 9.84 s · OL unavailable: no OpenLoops ls library serves g g > z z z z (public channel database of OpenLoops 2.1.5, 2026-11-06); row ML/RC only (PLAN §3.2) no verdict: TIRust failed closed 4/128 (gap) on TIRust · NinjaRS per amplitude: 16/128 failed closed, lane status Unstable
$2 \to 5$		
$gg \to e^+e^-\mu^+\mu^-g$ or $gg \to hgggg$ 7 legs · QCD · LI-51 OL 5.2e−12 · ML 6.6e−12 · pass · 3 pts	**112 s** ×89 cold 110 s · 2.88 GiB peak warm 2.88 GiB peak · plan 0.0530 MB Rc 1.26 s	**49.8 ms** ×0.14 Rc 116 ms · **OL 344 ms** · ML 777 ms $B = 128$ · 3 runtimes ×1.1 Rc not matched (values excluded): ×0.43 to it pass ×0.11 on TIRust **TIRust** (tirust) numerator 44 % vs 256-bit truth (16 pts): max 8.3e−9 **NinjaRS per amplitude** (ninjars) 209 ms ×0.61 NinjaRS 78 % vs 256-bit truth (16 pts): max 7.0e−9
EFT		
rank > N+2 EFT row (HC_NLO_X0 or SMEFT) EFT · LI-X1 · TIRust only · delegated (lit-eft)	***not timed*** execution check only (user 2026-11-06 ~14:55): rank > N+2 has no correct R2 (no dim-8 model), so no reference, no benchmark and no timing claim; lit-eft ran it (two dim-6 insertions through TIRust) no reference provider	



Not drawn: 2 of the 42 rows of Table 2 (2 no pyAmpliCol time at B = 128). Drawn with a provider that is not matched (its values no reference; no limit from it): $gg \to e^+e^-\mu^+\mu^-$ (Recola), $gg \to e^+e^-\mu^+\mu^-g$ (Recola).

Figure 1: **Batched full colour against the gate** (finite part, $B = 128$; the rows of Table 1 with a matched provider and, as diamonds, the rows of Table 2 with a pyAmpliCol time and a matched provider time).

Table 3: **TIRust and NinjaRS** (the two reduction backends) under the same public unchecked setting: time per point at $B = 128$ with each backend's stage bar (instrumented run), the ratio NinjaRS ÷ TIRust, and their agreement. Loop-induced rows: the timed rows of Table 2 (not shown: 1 row without an unchecked time, its marker in Table 2), each backend's largest relative error against TIRust at 256 bits (`data/li-unchecked-accuracy.json`) and the binary64 L6 verdict of the checked default (`lit.json`: status, TIRust's served points, the largest finite-part difference of the two backends). Born-interfered rows: Table 1's 2 → 3 and 2 → 4 rows, with the largest TIRust-vs-NinjaRS difference at the timed points.

Process	TIRust	NinjaRS	N ÷ T	T vs truth	N vs truth
Loop-induced ($\lvert A_{\text{loop}}\rvert^2$) · stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue					

Process	TIRust	NinjaRS	N ÷ T	T vs truth	N vs truth	
$gg \to h$	**0.345 ms** glue 85 %	**0.363 ms** glue 80 %	×1.1	1.2e−13	1.2e−12	pass (T 3/3) 3.2e−13; pass (T 3/3) 3.5e−13
$h \to gg$	**0.368 ms** glue 84 %	**0.386 ms** glue 80 %	×1.0	1.0e−13	1.1e−12	pass (T 3/3) 1.6e−13
$h \to aa$	**0.668 ms** inputs 70 %	**0.717 ms** inputs 74 %	×1.1	2.4e−14	4.2e−13	pass (T 3/3) 5.6e−14
$h \to za$	**1.000 ms** inputs 68 %	**1.06 ms** inputs 72 %	×1.1	1.2e−14	1.6e−12	pass (T 3/3) 2.8e−14
$aa \to h$	**0.708 ms** inputs 73 %	**0.742 ms** inputs 74 %	×1.0	1.4e−14	1.6e−13	pass (T 3/3) 5.5e−14
$gg \to aa$	**1.13 ms** numerator 47 %	**3.54 ms** NinjaRS 54 %	×3.1	6.6e−13	2.4e−13	pass (T 3/3) 1.6e−14; pass (T 3/3) 1.5e−14
$gg \to zz$	**2.36 ms** numerator 52 %	**6.10 ms** NinjaRS 61 %	×2.6	5.5e−12	8.8e−11	pass (T 3/3) 4.2e−12
$gg \to w^+w^-$	**1.82 ms** numerator 46 %	**8.06 ms** NinjaRS 66 %	×4.4	1.2e−12	5.2e−12	pass (T 3/3) 1.7e−13
$gg \to za$	**1.34 ms** numerator 49 %	**3.82 ms** NinjaRS 60 %	×2.9	6.8e−13	6.4e−12	pass (T 3/3) 1.6e−14
$gg \to hh$	**0.387 ms** glue 48 %	**0.564 ms** NinjaRS 41 %	×1.5	3.4e−9	8.8e−11	pass (T 3/3) 3.6e−14
$gg \to hz$	**0.729 ms** glue 37 %	**1.26 ms** NinjaRS 35 %	×1.7	1.4e−12	1.4e−12	pass (T 3/3) 8.3e−14
$gg \to hg$	**1.02 ms** glue 44 %	**1.66 ms** NinjaRS 44 %	×1.6	2.4e−14	7.3e−13	pass (T 3/3) 5.4e−16
$u\bar{u} \to hg$	**0.487 ms** glue 86 %	**0.532 ms** glue 81 %	×1.1	1.1e−14	1.5e−13	pass (T 3/3) 1.8e−15
$ug \to hu$	**0.297 ms** glue 84 %	**0.315 ms** glue 84 %	×1.1	1.2e−14	1.1e−13	pass (T 3/3) 3.0e−15
$gg \to zg$	**3.40 ms** glue 42 %	**12.9 ms** NinjaRS 66 %	×3.8	9.4e−13	9.7e−12	pass (T 3/3) 9.2e−14
$gg \to ag$	**1.15 ms** numerator 47 %	**3.09 ms** NinjaRS 65 %	×2.7	2.3e−13	1.7e−12	pass (T 3/3) 1.1e−13
$e^+e^- \to ha$	**1.86 ms** inputs 73 %	**1.98 ms** inputs 75 %	×1.1	3.8e−14	1.9e−13	pass (T 3/3) 9.5e−15
$aa \to aa$	**1.99 ms** inputs 63 %	**6.78 ms** inputs 62 %	×3.4	1.8e−14	4.9e−14	pass (T 3/3) 6.6e−15
$aa \to zz$	**4.28 ms** inputs 58 %	**11.8 ms** inputs 57 %	×2.8	1.5e−13	7.4e−13	pass (T 3/3) 4.9e−15
$aa \to hh$	**1.10 ms** inputs 74 %	**1.55 ms** inputs 74 %	×1.4	8.8e−11	9.1e−11	pass (T 3/3) 2.1e−14
$e^+e^- \to hh$	**1.00 ms** inputs 81 %	**1.02 ms** inputs 80 %	×1.0	1.3e−10	1.3e−10	pass (T 3/3) 1.2e−14

Process	TIRust	NinjaRS	N ÷ T	T vs truth	N vs truth	
$h \to ggg$	**1.10 ms** glue 40 %	**1.33 ms** NinjaRS 44 %	×1.2	6.7e−13	1.5e−12	pass (T 3/3) 2.4e−13
$z \to ggg$	**4.17 ms** numerator 47 %	**11.1 ms** NinjaRS 64 %	×2.7	5.5e−11	3.8e−10	pass (T 3/3) 7.4e−12
$e^+e^- \to ggg$	**5.21 ms** glue 41 %	**9.98 ms** NinjaRS 59 %	×1.9	3.5e−7	3.2e−10	partial (T 2/3) 9.6e−8
$gg \to hhh$	**4.60 ms** TIR moments 65 %	**5.36 ms** NinjaRS 57 %	×1.2	7.6e−7	**4.5e64 (2/128 > 1.0e−5)**	pass (T 3/3) 6.6e−13
$gg \to hgg$	**11.2 ms** numerator 43 %	**22.5 ms** NinjaRS 63 %	×2.0	9.2e−10	8.0e−10	pass (T 3/3) 2.7e−10
$gg \to zzg$	**76.0 ms** numerator 45 %	**460 ms** NinjaRS 82 %	×6.1	**1.2e−2 (18/128 > 1.0e−5)**	5.0e−8	partial (T 2/3) 3.2e−10
$gg \to aag$	**24.4 ms** numerator 58 %	**170 ms** NinjaRS 79 %	×7.0	7.0e−6	**1.0e−3 (1/128 > 1.0e−5)**	partial (T 0/3)
$gg \to w^+w^-g$	**65.0 ms** TIR moments 43 %	**320 ms** NinjaRS 81 %	×4.9	**1.4e−2 (24/128 > 1.0e−5)**	6.2e−8	partial (T 0/3)
$gg \to zzz$	**43.3 ms** numerator 68 %	failed closed: 20/128 failed closed, lane status Unstable	—	2.3e−8	—	pass (T 3/3) 5.3e−11
$gg \to hzg$	**14.2 ms** numerator 54 %	failed closed: 2/128 failed closed, lane status Unstable, lane status MasterFailed	—	8.2e−9	—	pass (T 3/3) 3.8e−13
$gg \to agg$	**23.1 ms** numerator 54 %	**98.3 ms** NinjaRS 78 %	×4.3	3.3e−6	**2.0e−4 (1/128 > 1.0e−5)**	partial (T 0/3)
$gg \to zgg$	**72.7 ms** numerator 41 %	**330 ms** NinjaRS 78 %	×4.5	**3.1e−5 (3/128 > 1.0e−5)**	6.2e−6	partial (T 0/3)
$u\bar{u} \to hgg$	**2.92 ms** TIR moments 40 %	**2.39 ms** NinjaRS 47 %	×0.82	3.7e−9	3.5e−8	partial (T 2/3) 3.2e−11
$u\bar{u} \to d\bar{d}h$	**2.46 ms** glue 91 %	**0.991 ms** glue 89 %	×0.40	4.1e−10	4.3e−9	pass (T 3/3) 1.3e−10
$gg \to hhgg$	**331 ms** TIR moments 78 %	**600 ms** NinjaRS 87 %	×1.8	7.1e−7	7.0e−7	partial (T 1/3) 3.0e−10; partial (T 0/1)
$gg \to hggg$	**257 ms** numerator 43 %	**950 ms** NinjaRS 75 %	×3.7	5.9e−8	**4.2e−5 (2/32 > 1.0e−5)**	partial (T 0/3)
$gg \to e^+e^-\mu^+\mu^-$	**3.57 ms** glue 38 %	**5.58 ms** NinjaRS 40 %	×1.6	5.4e−12	6.5e−10	pass (T 3/3) 2.6e−13
$gg \to aagg$	**720 ms** numerator 45 %	**5.50 s**	×7.6	**T vs N 7.0e−2 (23/128 > 1.0e−5) (no truth)**	**T vs N 7.0e−2 (23/128 > 1.0e−5) (no truth)**	partial (T 0/3)
$gg \to zzzz$	failed closed: 4/128 failed closed, lane status TIRust Unstable	failed closed: 16/128 failed closed, lane status Unstable	—	—	—	partial (T 0/3)
$gg \to e^+e^-\mu^+\mu^-g$	**49.8 ms** numerator 44 %	**209 ms** NinjaRS 78 %	×4.2	8.3e−9	7.0e−9	pass (T 3/3) 6.9e−12
Born-interfered (Table 1's G1 rows) · LI-B1's L6 over its campaign rows: partial (85 pass, 7 not run, 115 partial, 9 pass (lax))						
$u\bar{u} \to d\bar{d}g$	**2.41 ms**	**2.29 ms**	×0.95	largest |T − N| ÷ max(|T|, |N|) 1.4e−9 at 128 points		
$u\bar{u} \to u\bar{u}g$	**2.56 ms**	**3.45 ms**	×1.3	largest |T − N| ÷ max(|T|, |N|) 4.7e−6 at 128 points		
$u\bar{u} \to e^-e^+g$	**0.639 ms**	**0.610 ms**	×0.95	largest |T − N| ÷ max(|T|, |N|) 8.2e−12 at 128 points		
$u\bar{u} \to ggg$	**9.92 ms**	**6.47 ms**	×0.65	largest |T − N| ÷ max(|T|, |N|) 1.8e−9 at 128 points		
$gg \to ggg$	**32.6 ms**	**25.9 ms**	×0.79	largest |T − N| ÷ max(|T|, |N|) 2.1e−3 at 128 points		
$u\bar{u} \to e^-e^+gg$	**11.3 ms**	**4.94 ms**	×0.44	largest |T − N| ÷ max(|T|, |N|) 2.1e−8 at 128 points		
$gg \to t\bar{t}gg$	**410 ms**	**129 ms**	×0.31	largest |T − N| ÷ max(|T|, |N|) 2.7e−8 at 128 points		
$u\bar{u} \to gggg$	**286 ms**	**81.7 ms**	×0.29	largest |T − N| ÷ max(|T|, |N|) 5.6e−6 at 128 points		
$gg \to gggg$	**1.33 s**	**621 ms**	×0.47	largest |T − N| ÷ max(|T|, |N|) 1.8e−3 at 128 points		

Missing or incomplete (generated from the tables' data; Table 1 then Table 2; gate lines: time ÷ limit and the stage that dominates pyAmpliCol's time)
T1 · no value: leading colour: not supported (refused by design: EW LC/NLC is never supported (user 2026-10-30, Q4) — $gg \to t\bar{t}e^-e^+g$
T1 · no value: no matched provider time (no gate) — $gg \to ggggg$
T2 · no value: time: failed closed (4/128 failed closed, lane status TIRust Unstable) — $gg \to zzzz$
T2 · no value: not timed: execution check only (user 2026-11-06 ~14:55): rank > N+2 has no correct R2 (no dim-8 model), so no reference, no benchmark and no timing claim — rank > N+2 EFT row (HC_NLO_X0 or SMEFT)
T2 · gate: open on TIRust — $e^+e^- \to ggg$ (×0.90, glue 41 %), $gg \to hhgg$ (×1.3, TIR moments 78 %)
T2 · validation: partial: L3, L4, L5 (checked default; lit.json) — $h \to ggg$, $z \to ggg$, $gg \to zzz$
T2 · validation: partial: L3, L4, L5, L6; TIRust fails 1 of 3 binary64 fixture points closed (checked default; lit.json) — $e^+e^- \to ggg$, $gg \to zzg$, $u\bar{u} \to hgg$
T2 · validation: partial: L3, L4, L5, L6 (checked default; lit.json) — $gg \to hgg$
T2 · validation: partial: L3, L4, L5, L6; TIRust fails 3 of 3 binary64 fixture points closed (checked default; lit.json) — $gg \to aag$, $gg \to w^+w^-g$, $gg \to agg$, $gg \to zgg$, $gg \to hggg$, $gg \to aagg$, $gg \to zzzz$
T2 · validation: partial: L3, L4, L5, L6; TIRust fails 2 of 3, 1 of 1 binary64 fixture points closed (checked default; lit.json) — $gg \to hhgg$
T2 · accuracy: NinjaRS per amplitude unchecked: 2/128 points above 1e-05, worst 4.5e+64 — $gg \to hhh$
T2 · accuracy: TIRust unchecked: 18/128 points above 1e-05, worst 1.2e-02 — $gg \to zzg$
T2 · accuracy: NinjaRS per amplitude unchecked: 1/128 points above 1e-05, worst 1.0e-03 — $gg \to aag$
T2 · accuracy: TIRust unchecked: 24/128 points above 1e-05, worst 1.4e-02 — $gg \to w^+w^-g$
T2 · accuracy: NinjaRS per amplitude unchecked: 20 points fail (Unstable) — $gg \to zzz$
T2 · accuracy: NinjaRS per amplitude unchecked: 2 points fail (MasterFailed, Unstable) — $gg \to hzg$
T2 · accuracy: NinjaRS per amplitude unchecked: 1/128 points above 1e-05, worst 2.0e-04 — $gg \to agg$
T2 · accuracy: TIRust unchecked: 3/128 points above 1e-05, worst 3.1e-05 — $gg \to zgg$
T2 · accuracy: NinjaRS per amplitude unchecked: 2/32 points above 1e-05, worst 4.2e-05 — $gg \to hggg$
T2 · accuracy: TIRust vs NinjaRS unchecked (no 256-bit truth, cross-backend, not a truth): 23/128 points differ above 1e-05, largest 7.0e-02; at its 2 largest-difference points against the 256-bit truth: TIRust 1.3e-04, NinjaRS 7.6e-02 — $gg \to aagg$

Table 4: **Gate and stage split at $B = 128$**, ms per point: the numbers behind the gate chips and stage bars of Table 1 and behind Figure 1. pyAmpliCol's median (full colour), the binding limit and its provider, the uncertainty band of time ÷ limit, the gate chip of Table 1, and pyAmpliCol's time per stage.

Process	ms/pt	Limit	Band	Gate	inputs	numerator	scatter	NinjaRS	masters	glue
$gg \to gg$	**1.44**	—	—	no gate (2→2)	0.241	0.573	0.0871	0.162	0.0527	0.229
$u\bar{u} \to d\bar{d}g$	**3.02**	4.69 OL	0.590–0.698	<5 ms: not gated ×0.64	0.171	1.01	0.657	0.327	0.218	0.614
$u\bar{u} \to u\bar{u}g$	**2.43**	4.96 OL	0.468–0.513	<5 ms: not gated ×0.49	0.158	0.922	0.458	0.290	0.204	0.402
$u\bar{u} \to e^-e^+g$	**0.672**	0.770 Rc	0.858–0.887	<5 ms: not gated ×0.87	0.0492	0.227	0.131	0.0344	0.0236	0.193
$u\bar{u} \to ggg$	**6.73**	29.1 OL	0.202–0.260	pass ×0.23	0.997	2.84	0.668	0.931	0.354	0.871
$gg \to ggg$	**24.6**	222 OL	0.0979–0.123	pass ×0.11	7.53	10.0	0.774	4.49	0.784	1.44
$u\bar{u} \to e^-e^+gg$	**6.14**	24.5 Rc	0.232–0.268	pass ×0.25	0.256	2.57	0.804	0.690	0.330	0.679
$gg \to t\bar{t}gg$	**76.1**	652 OL	0.116–0.118	pass ×0.12	12.8	27.7	1.01	21.6	7.01	5.52
$u\bar{u} \to gggg$	**58.7**	517 OL	0.109–0.118	pass ×0.11	11.4	27.9	1.85	10.2	3.17	3.17
$gg \to gggg$	**463**	6 240 OL	0.0653–0.0832	pass ×0.074	188	165	3.29	72.0	8.68	7.47
$u\bar{u} \to e^-e^+d\bar{d}g$	**14.3**	49.7 Rc	0.274–0.300	pass ×0.29	0.840	6.35	2.00	2.43	1.60	1.39
$u\bar{u} \to e^-e^+ggg$	**43.2**	372 Rc	0.115–0.117	pass ×0.12	1.91	24.9	2.23	8.97	2.89	2.36
$gg \to t\bar{t}e^-e^+g$	**75.4**	944 OL	0.0653–0.0945	pass ×0.080	5.81	33.4	1.18	20.4	6.20	3.06
$gg \to t\bar{t}e^-e^+g$ (EW)	**181**	1 980 OL	0.0854–0.0973	pass ×0.091	12.1	66.9	2.69	54.8	22.6	10.8
$u\bar{u} \to ggggg$	**1 810**	569 000 ML	0.00219–0.00415	pass ×0.0032	367	913	12.3	364	63.7	36.7
$gg \to ggggg$	**5 110**	—	—	no gate (no matched provider)	2 340	2 250	10.3	444	44.7	30.5

Reading the tables

Timing settings (every time in Table 2, Table 3 and Figure 1's loop-induced rows: most optimistic, no stability check, reached through settings a user sets; nothing modified; Table 1 as stated below; the stored $2 \to 6$ rows of Table 1 are outside this protocol: internal checks and rescue, labelled in their rows). **pyAmpliCol** (binary64, SymJIT `-O2` evaluators as shipped, `RAYON_NUM_THREADS=1`, one core by `taskset`): Table 2 and Table 3 time `Runtime.load(artifact, reduction_unchecked=True)` then `evaluate(points, mu_r2=…, poles="none", on_failure="return", reduction="tirust" | "ninjars")`, with no `stability_checks`, no `stability` rescue policy and `kinematics_improvement_binary64` off (the defaults); Table 1 times the same public path through the benchmark harness (`python -m tools.loop_benchmark pyamplicol <row> --inputs native --batches 128`, values bitwise those of `Runtime.evaluate`), poles off, no stability check or rescue, the default reduction setting (`reduction_unchecked = false`, NinjaRS: its per-point acceptance test runs, with no rescue behind it). On the nine $2 \to 3$ and $2 \to 4$ rows a paired same-core A/B against `reduction_unchecked = True` (3 fresh instances per arm, 128 points) gives bitwise-identical values and unchecked ÷ default ×0.92–×1.25 (×1.25 a noisy g g ⟶ g g g sample; the others ×0.92–×1.05): these times equal the no-check setup within the timing noise. **OpenLoops** 2.1.5: `expert_mode 1`, `stability_mode 11` (stability system off), `truepoles 0`, and `ir_on 0` on the Born-interfered rows. **MadLoop** (MG5_aMC 3.7.0): `CTModeRun 1` in `MadLoopParams.dat` (no stability test). **Recola** 2.2.5 with COLLIER: `set_dynamic_settings_rcl(0)`, `set_compute_ir_poles_rcl(0)`, release build. A provider time marked [P] is a production record (named in its table's legend).

Table 1 (Born-interfered one-loop processes) has one row per test process, by multiplicity. A cell gives pyAmpliCol's value in bold and, at its right, the multiplier *pyAmpliCol ÷ the fastest provider* on this host (in bold among the provider values). A provider time marked ≤ ($u\bar{u} \to ggggg$) comes from a run outside the protocol that does the production work and more (Recola's reference run evaluates the IR poles too), so it bounds that provider's production time from above, and the multiplier against it is a lower bound (≥); it is never a gate limit. Only run times are coloured: green faster, white at parity, red slower ($\log_2$ scale, clamped at ×1/16 and ×16). Generation multipliers are uncoloured (generation time is not the main metric) and leading colour (LC) has no provider ratio. Grey

cells say why a value is missing. Run times are medians per phase-space point at $B = 128$, finite part, binary64, one pinned thread, no stability check or rescue, the default reduction setting (`reduction_unchecked = false`: NinjaRS's per-point acceptance test runs; see *Timing settings*); generation is generate + plan export (model loading excluded) with its peak memory and plan size. The providers run in their most optimistic settings (m-lit F4; `tools/loop_benchmark providers --protocol optimistic`), timed on pyAmpliCol's core: MadLoop 3.7.0 (ML) without its stability test (`CTModeRun 1`), Recola 2.2.5 with COLLIER (Rc) without dynamic settings (alpha_s and the scale set before its generation, outside the timer, so it times the stream points at the first point's μ_R: 6 points per row, against up to 16 for the other providers), `compute_ir_poles 0`, release build, OpenLoops 2.1.5 from prebuilt libraries (OL: no generation time) with its stability system and I-operator poles off; each optimistic value equals the provider's production value at the timed points. A provider time marked [p] is its production record, not re-timed (gg_gggg recola (not re-run); uux_ggggg madloop (its recorded generation / initial call 6006 s: beyond the 45-minute rule, HANDOFF 0.1); uux_ggggg recola (its recorded generation / initial call 10220 s: beyond the 45-minute rule, HANDOFF 0.1)). Each cell shows the most recent measured configuration (`scripts/build_summary.py`, by git ancestry of the revisions), named once in the table's first row (engine, native build and digest, date, host); a cell from another configuration names its origin.

Gate chips (Table 1, Figure 1, Table 4): the time over the binding limit, the smallest of MadLoop's time and 1.30 × each of OpenLoops' and Recola's; **pass** or **fail** when the whole uncertainty band (Table 4) lies below or above 1, else **open**. A process below 5 ms per point has no blocking gate ("<5 ms: not gated", user rule of 2026-10-24) and its chip keeps the measured ratio; $2 \to 2$ has no gate. The gate is taken at $B = 128$ only.

Stage bars (every bar of the report, one colour per stage; the cell names the largest share; Table 4 gives every stage of Table 1 in ms): inputs = trees, Born and counterterm inputs (loop-induced: the trees and counterterm kets of the amplitude basis); numerator = seeds, fused factors and the dressing of the open-loop objects; scatter = closures into the reduction lanes; then the reduction backend's own stages: NinjaRS = its expansions (the reduction without the masters) and masters = the master integrals (canonical arguments, cache, oneloopmaster), or TIR moments = TIRust's tensor moments and contraction = their contraction with the numerator (TIRust's master integrals, a part of its tensor moments, are shown apart only where a profiled run recorded them); glue = point parameters, input conversion, transport, squaring, assembly and the rest of the call.

Validation (process cell): the largest deviation $|a - b| \,/\, \max(|a|, |b|, |B|\ \alpha_s/2\pi)$ of the binary64 finite part per reference at the timed points (NJet DD: NJet 3.1.2 in double-double), and the median over the points and references; † the largest lies within pyAmpliCol's binary64 stability estimate at that point; ¶ it is the reference's own: pyAmpliCol's double-double value lies as far from it, and its binary64 value lies within its estimate of the double-double one. ‡ $gg \to ggggg$: the reference check's points 4 and 8 (1.5×10^{-7} and 2.6×10^{-8} from NJet DD) exceed its tolerance 2×10^{-8} in binary64: **accepted (user 2026-10-25): binary64 within the estimate, DD = NJet** (estimates E 1.5×10^{-6} and 3.4×10^{-7}; pyAmpliCol's double-double values equal the reference's); the check keeps its tolerance and, asserting double-double at these points, passes.

Notes under a value qualify it and leave it as measured, in amber a known limitation, in grey its run conditions: $gg \to t\bar{t}e^-e^+g$, full colour ("model: the workspace's py3 G_mu copy"): timed on reference/models/loop_qcd_qed_sm_Gmu_py3 (the harness's processes.json default, as at M-SM), not on the shipped patched UFO (UFOmodeldb 6ca4b120 + the user's issue-1 patch) that the M-SM campaign validated; the speed gate is taken on this timing.

The $2 \to 6$ rows (`summary.json preliminary`, from `data/2to6.json`) have no provider, so no multiplier, gate or colour; their engine is stated once in the table. Generation = generate + plan export, with the store, artifact and load beside it ("FC" marks the full-colour generation, LC below it); the timed evaluation is binary64, finite part, checks off, one pinned thread, at the largest batch measured. The process cell lists the internal checks (ε^{-2}: double pole against the universal value; E: the stability estimate, with the points above 10^{-5}; rescue: the stream points the default rescue policy's binary64 level sends up; Ward on the external gluons; binary64 against double-double; Catani at full colour; NGluon at leading colour on the all-gluon row) and, in grey, those not run or not asserted.

Table 2 follows Table 1′s rules for loop-induced processes, $|A_{\text{loop}}|^2$ without a Born: one row per process of the M-LIT validation matrix (`m-lit/PLAN.md` §4.1; its ids in the process cell), grouped by multiplicity; the backend-unit, zero, identity, observable and complex-mass-scheme rows (LI-B1, LI-B2, LI-Z1, LI-Z1b, LI-Z2, LI-W1, LI-L1, LI-P1, LI-C1, LI-S1, LI-O1) are validation-only and not listed. On every timed row, the cell is the batched $B = 128$ time per point of TIRust with the **unchecked reduction setting** (`reduction_unchecked`, a documented public setting, off by default): each backend reduces without its own acceptance or refusal checks. TIRust keeps its a priori conditioning gates and its method choice as method selection, computes no acceptance pass, and serves what the method choice refuses by plain Passarino–Veltman (gate 0); NinjaRS runs with zero stability thresholds; pyAmpliCol's own checks are off. Only what no scheme evaluates finitely (an exactly singular pivot, a non-finite output, a master failure, a NinjaRS lane whose tested quantity is exactly zero or NaN, `Unstable`) still fails. The time is the median over 3 fresh runtimes (created in rotated order) of each one's median over rounds of the public call (`Runtime.load` + `evaluate` on the 128 distinct stream points, poles off, one pinned thread); the spread (×, largest over smallest runtime) is beside B; `auto` with the setting is TIRust, bitwise and without fallback (paired on gg_hgg). Beneath, NinjaRS with the same setting: its time and multiplier where it served every point, else the points it failed with their cause (a call that fails points is timed whole: no served-point time). The coloured bars are the stage shares (the stages of Table 1′s bars, the reduction split into each backend's own stages) of a **separate instrumented run** with the same setting, points and B on the same core, never the timed call. The setting has **no accuracy certificate**: it is a timing and benchmark mode. Beneath each backend's line, its measured accuracy on the row's first stream points (the same public call, binary64 finite part) against TIRust at 256 bits (`data/li-unchecked-accuracy.json`): the largest relative error, or in red the points above 10^{-5} with

the worst and how they were served (TIRust: by the gate-0 PV last resort; NinjaRS: values the checked mode refuses, or also in the checked mode) and the points that fail. The providers run in their most optimistic settings (m-lit F4: OpenLoops with its stability system off, MadLoop without its stability test, Recola without dynamic settings, in its release build), in the same unit on the same core as pyAmpliCol, alternating with it between rounds, their time the median over those rounds; the multiplier against the fastest matched provider colours the cell with Table 1′s scale. A provider whose timing record excludes its values as a reference (Recola's on-shell widths against the fixed-width scheme of the others on the rows LI-43 and LI-51) is timed and shown but is no matched provider (its values miss the 1e-5 agreement of `m-lit/PLAN.md` §4.0 L1; §4.1 names MadLoop and OpenLoops as the references of LI-43 and OpenLoops of LI-51): never the fastest matched provider and never a limit, the cell gives the multiplier to it, uncoloured. Generation, uncoloured, is the engine's generation (model loading excluded): generate + the exact helicity-zero filter + plan export, the multiplier formed on it. The generation-time finiteness, Ward and Goldstone probes are validation, off by default (`lit-probe`; the user enables them, in binary64 with a double-double fallback): where a record ran them their time is shown apart beneath the value with its carrier (every record before `lit-probe` ran them in double-double), never in it; the peak memory is over all stages; the provenance line names pyAmpliCol's host and the providers'. Rows of four legs or fewer ($2 \to 1$, $1 \to 2$, $2 \to 2$, $1 \to 3$) are reported without a gate; from $2 \to 3$ upward the gate is Table 1′s (faster than MadLoop and at most 1.30× each matched OpenLoops and Recola time, on TIRust), "<5 ms: not gated" below 5 ms per point, no gate without a matched provider; verdict chips come with the gate tool's records and are TIRust's (the gated backend, "on TIRust"); where TIRust failed points closed the chip says "no verdict … (gap)": its time is no served-point time and how a verdict is given there is not ruled ("no verdict yet" before the gate tool's records, "no provider time yet" while no provider of the row is timed). Grey cells are explicit markers: **not run** (a landed package without a record of the row, or a run stopped before its timing, with its measured progress, e.g. a generation beyond the 45-min rule), **failed closed** or **refused** (the record of the cell's request failed points or was refused: no time), and a provider's reason where it has no matched time. Nothing is estimated or extrapolated; each cell shows the newest record of its row. At this head 42 of 42 rows are served: 40 timed, 0 not run inputs: `data/li-timing.json` present, `tools/loop_reference/li_campaign.json (provider_timing_records)` present, `data/lit.json` present; the registry against `lit-ref`'s manifest (`tools/loop_reference/li_campaign.json`): checked (39 rows, 42 ids).

Figure 1 places each row's batched full-colour value of Table 1 (dots) against the providers (diamonds, the darkest the fastest) and the gate limit (red bar, the shaded side fails; grey bar: the limit of a process below 5 ms, not gated; dotted line: 5 ms), for the rows with a matched provider. Beneath them, under a rule, the rows of Table 2 with a batched pyAmpliCol time and a matched provider time, pyAmpliCol as a larger diamond in its colour, on the same axis with their providers and, from $2 \to 3$ upward, the gate limit (MadLoop's time and 1.30× each other matched provider's, the smallest); the rows without both are not drawn, counted in the key by reason.

Table 4 lists, for every row of Table 1 timed at $B = 128$, the numbers its gate chip and stage bar summarise: the binding limit and its provider, the uncertainty band of time ÷ limit (the standard error of pyAmpliCol's median over its timing blocks plus the provider's run-to-run half-range), and the time of each stage. Data: `data/summary.json` and `data/summary-li.json`, written by `scripts/build_summary.py` from the campaign records in `data/` (`README.md`).